

SMART TEMPERATURE AND HUMIDITY MONITORING SYSTEM USING ARDUINO

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ABSTRACT

The Smart Temperature and Humidity Monitoring System is developed using Arduino Uno, DHT22 sensor, 16×2 I2C LCD display, and a buzzer. The system continuously monitors temperature and humidity and displays the readings on the LCD. When the temperature exceeds 30°C, the buzzer generates an audible alert.

INTRODUCTION

This project provides a simple and cost-effective solution for monitoring temperature and humidity in real time using Arduino technology.

OBJECTIVES

- Measure temperature and humidity using DHT22 sensor.
- Display sensor readings on LCD.
- Generate an alert when temperature exceeds 30°C.
- Learn sensor interfacing and embedded programming.

PROJECT IMPLEMENTATION

Figure 1: Entire Circuit

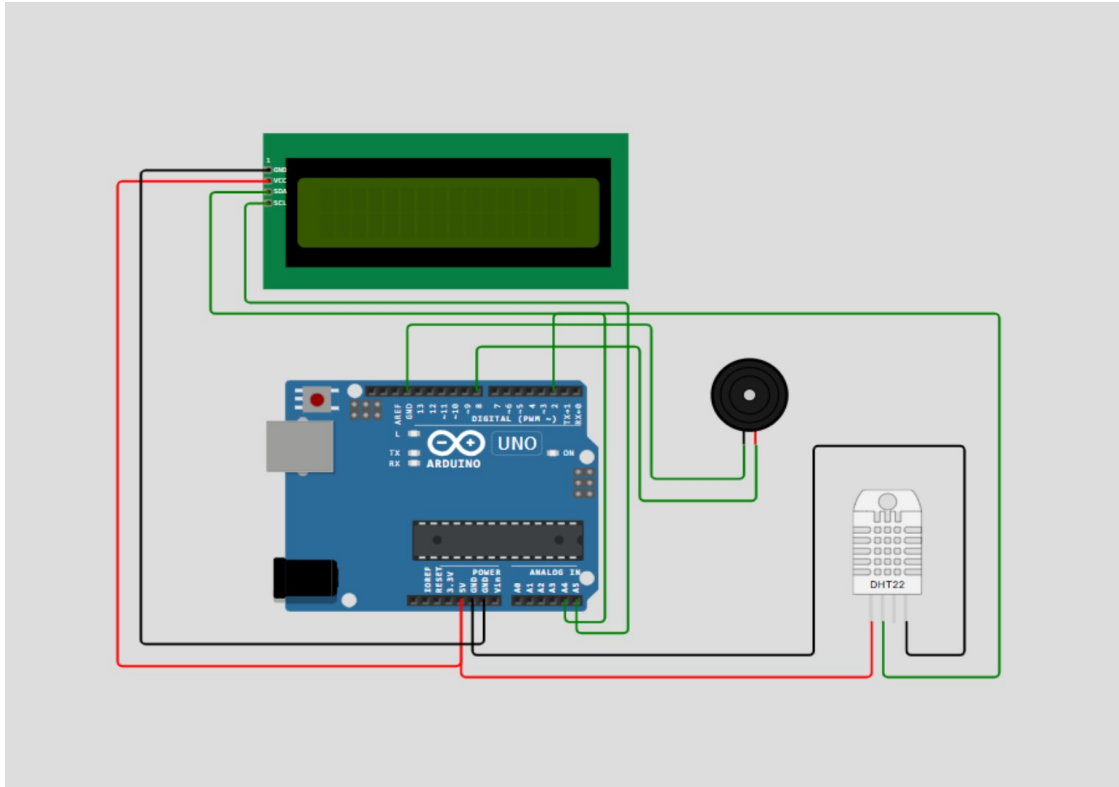


Figure 1 shows the complete circuit diagram of the Smart Temperature and Humidity Monitoring System. It consists of an Arduino Uno, DHT22 sensor, 16×2 I2C LCD display, and buzzer connected according to the circuit design.

Figure 2: Normal Condition

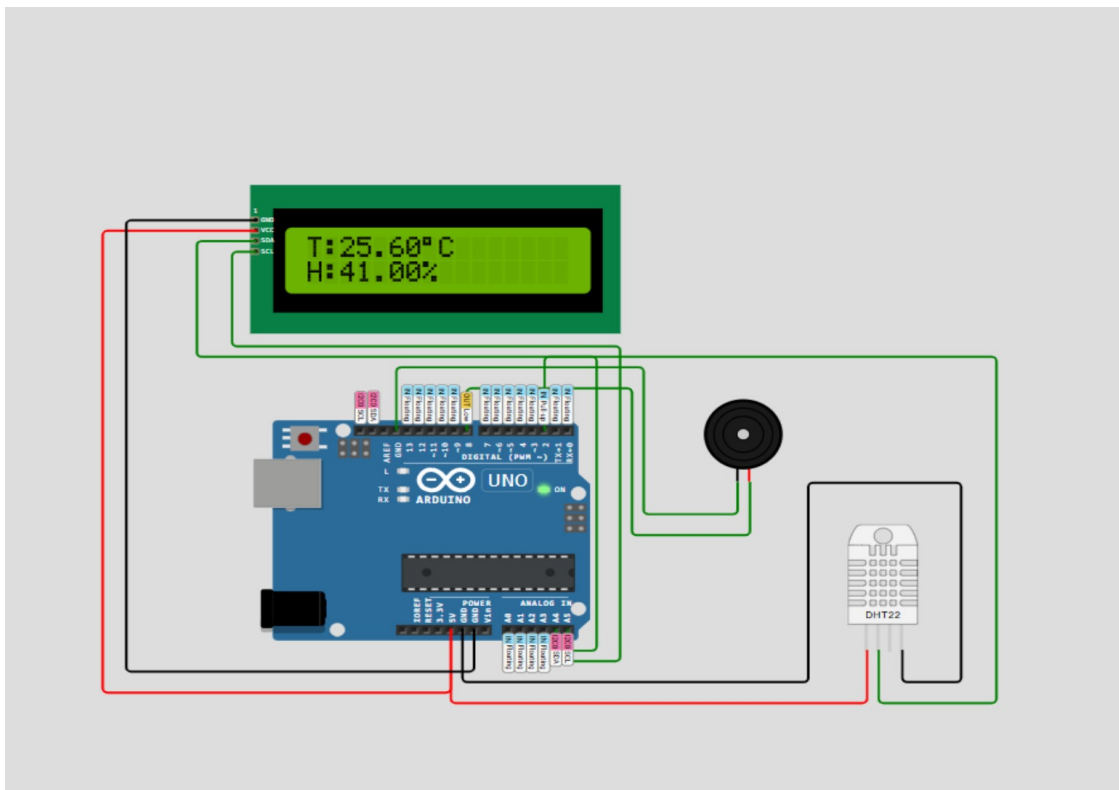


Figure 2 shows the system under normal operating condition. The temperature is below 30°C, the LCD displays the current temperature and humidity values, and the buzzer remains OFF.

Figure 3: Alert Condition

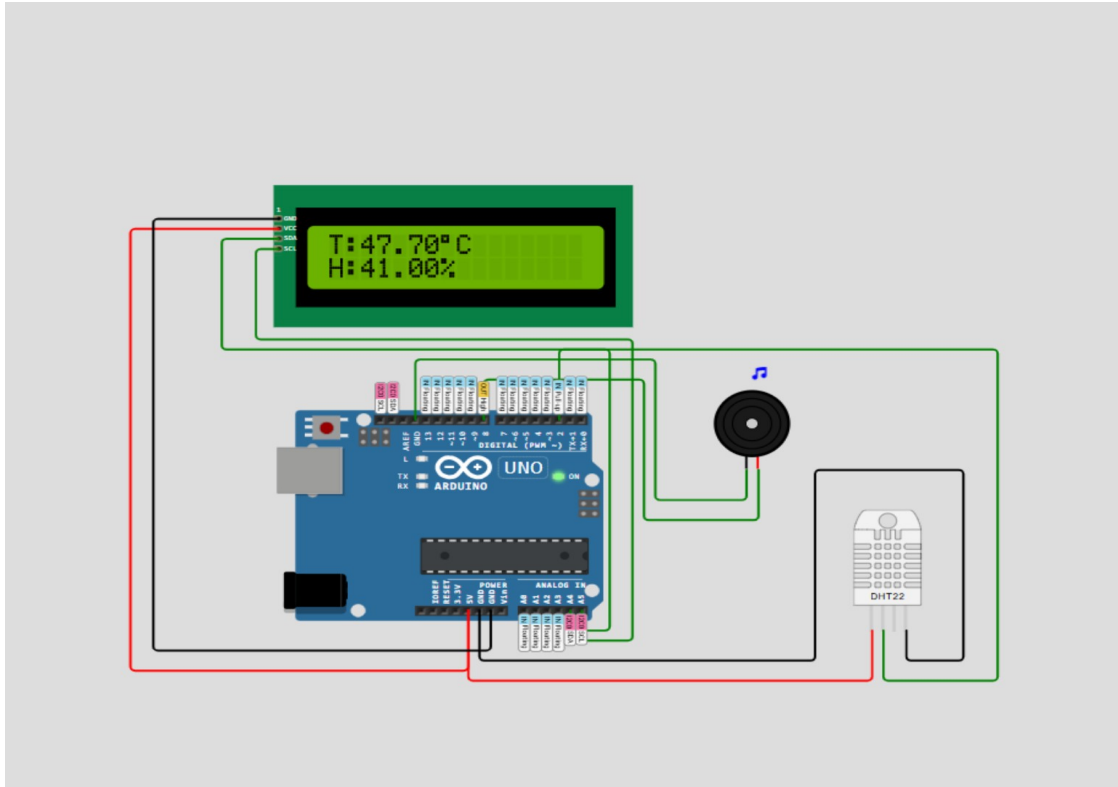


Figure 3 shows the alert condition of the system. When the temperature exceeds 30°C, the LCD displays the updated temperature and humidity values, and the buzzer turns ON to indicate a high-temperature alert.

RESULTS

The system successfully monitored temperature and humidity and generated alerts whenever the temperature exceeded 30°C.

APPLICATIONS

Smart Homes, Greenhouses, Laboratories, Warehouses, Industrial Monitoring Systems, Server Rooms.

CONCLUSION

The project demonstrates real-time environmental monitoring and automatic alert generation using embedded system technology.